

## 2008 Ford Edge SE

### 2008 BRAKES Hydraulic Brake Actuation - Edge & MKX

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## SPECIFICATIONS

### Material

| Item                                                                              | Specification               | Fill Capacity               |
|-----------------------------------------------------------------------------------|-----------------------------|-----------------------------|
| High Performance DOT 3 Motor Vehicle Brake Fluid<br>PM-1-C (US); CPM-1-C (Canada) | WSS-M6C62-A or WSS-M6C65-A1 | 803-821 ml (1.697-1.735 pt) |

## TORQUE SPECIFICATIONS

| Description                         | Nm | lb-ft |
|-------------------------------------|----|-------|
| Brake master cylinder nuts          | 25 | 18    |
| Brake pedal bracket nuts            | 22 | 16    |
| Master cylinder brake tube fittings | 28 | 21    |

## DESCRIPTION AND OPERATION

### HYDRAULIC BRAKE ACTUATION

The hydraulic brake actuation system consists of the following components:

- Brake master cylinder
- Brake fluid reservoir
- Brake pedal and bracket assembly
- Brake tubes and hoses
- Hydraulic control unit (HCU)

**Hydraulic Brake Actuation** - The brake pedal is connected to the power brake booster, which is connected to the brake master cylinder. When the brake pedal is pressed, brake fluid is forced from the master cylinder through the steel tubing and flexible hoses to the front disc brake calipers and the rear disc brake calipers. The front disc brake caliper pistons are forced outward against the brake pads, which contact the braking surface of the brake discs. The brake fluid calipers are pressurized by the master cylinder, forcing the disc brake caliper pistons outward against the brake pads. The brake pads press against the braking surface on the brake discs. When the brake pedal is released, the pressure is relieved, returning the brake caliper pistons and the brake pads to the at-rest position.

The dual brake system is split front and rear with the front wheel brakes comprising one circuit and the rear wheel brakes the other circuit.

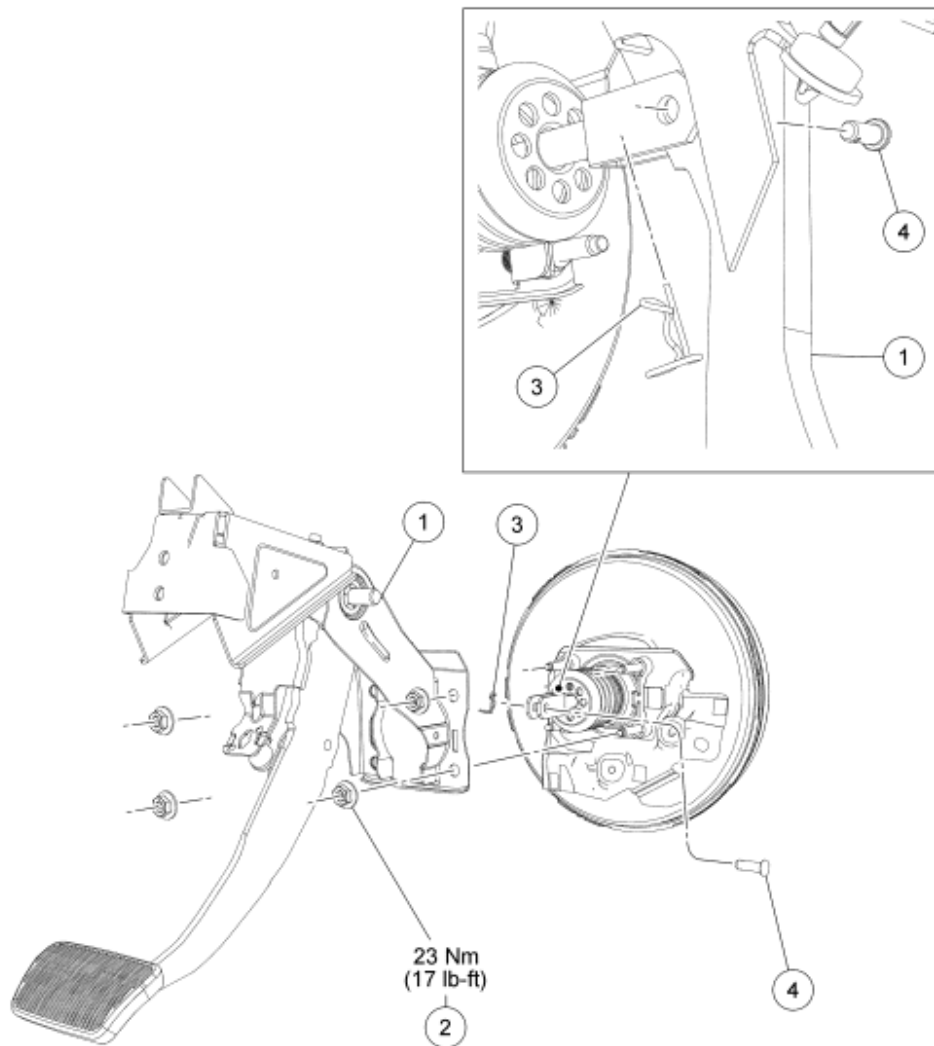
## DIAGNOSTIC TESTS

## HYDRAULIC BRAKE ACTUATION

Refer to **BRAKE SYSTEM - GENERAL INFORMATION** article.

## REMOVAL AND INSTALLATION

### BRAKE PEDAL AND BRACKET



N0076975

**Fig. 1: Identifying Brake Pedal and Bracket**  
Courtesy of FORD MOTOR CO.

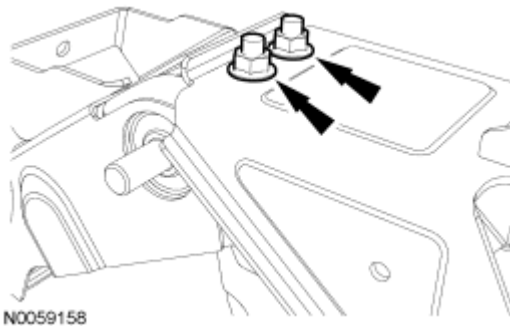
| Item | Part Number | Description                          |
|------|-------------|--------------------------------------|
| 1    | 2455        | Brake pedal and bracket              |
| 2    | W707144     | Brake pedal bracket nut (4 required) |
|      |             |                                      |

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|   |       |                                 |
|---|-------|---------------------------------|
| 3 | 7B096 | Brake booster push rod pin clip |
| 4 | 2N513 | Brake booster push rod pin      |

**CAUTION:** Do not remove or loosen the fasteners that hold the 2 brake pedal bracket halves together. Failure to follow these instructions may result in damage to the brake pedal bracket.

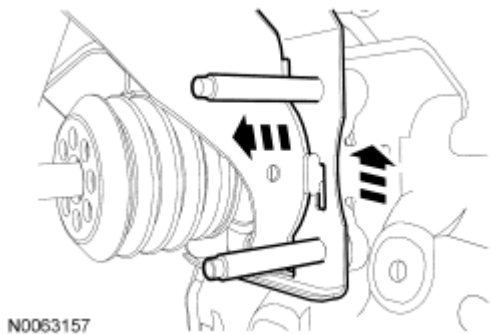


**Fig. 2: Locating Fasteners On Brake Pedal Bracket Halves**

#### REMOVAL AND INSTALLATION

**CAUTION:** Do not service the brake pedal or brake booster without first removing the stoplamp switch and speed control deactivator switch. These switches must be removed with the brake pedal in the at-rest position. Switch plungers must be compressed for the switch to rotate in the bracket. Attempting to remove the switch when the plunger is extended (during pedal apply) will result in damage to the switch.

1. Remove the stoplamp switch. For additional information, refer to **EXTERIOR LIGHTING** article.
2. Remove the instrument panel. For additional information, refer to **INSTRUMENT PANEL AND CONSOLE** article.
3. Remove the brake booster push rod pin clip and pin, then disconnect the push rod from the brake pedal arm.
4. Remove the 4 brake pedal bracket nuts.
  - To install, tighten to 22 Nm (16 lb-ft).
5. Lift the brake pedal and bracket off of the hangers and remove the brake pedal and bracket.



**Fig. 3: Lifting Brake Pedal & Bracket Off Of Hangers & Removing Brake Pedal & Bracket**  
Courtesy of FORD MOTOR CO.

**CAUTION:** Do not press, pull or otherwise move the brake pedal while installing the stoplamp switch or the speed control deactivator switch. These switches must be installed with the booster push rod attached to the brake pedal and with the brake pedal in the at-rest position. Installing these switches with the brake pedal in any other position will result in incorrect adjustment and may damage the switches.

6. To install, reverse the removal procedure.
- Apply brakes several times to verify correct brake operation.

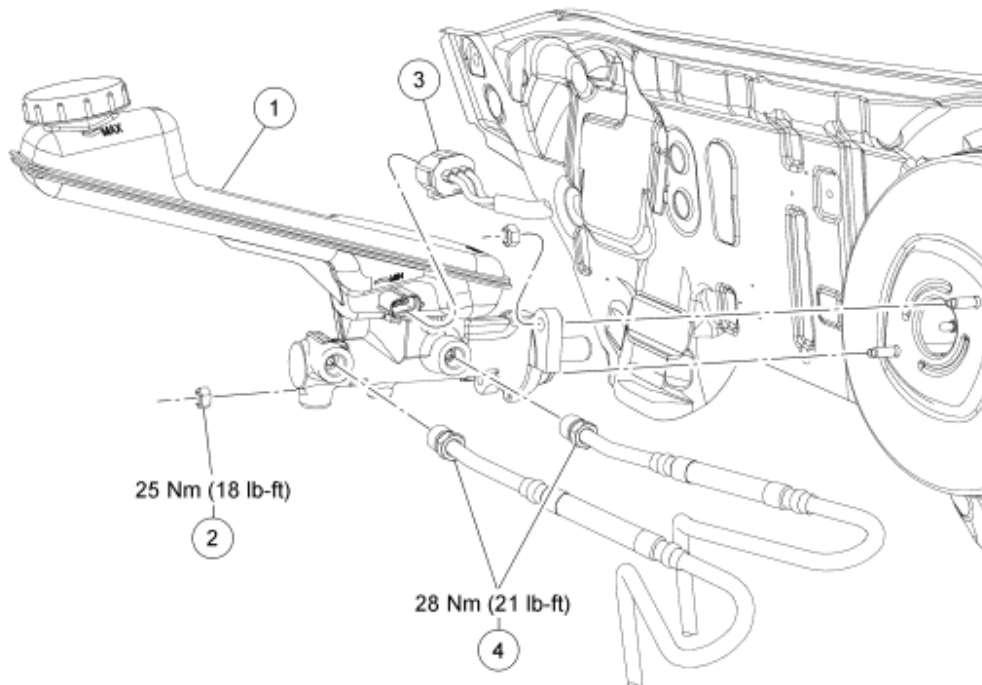
**BRAKE MASTER CYLINDER**

**Material**

| Item                                                                              | Specification               |
|-----------------------------------------------------------------------------------|-----------------------------|
| High Performance DOT 3 Motor Vehicle Brake Fluid<br>PM-1-C (US); CPM-1-C (Canada) | WSS-M6C62-A or WSS-M6C65-A1 |

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**Fig. 4: Exploded View Of Brake Master Cylinder With Torque Specifications**  
Courtesy of FORD MOTOR CO.

| Item | Part Number       | Description                                                    |
|------|-------------------|----------------------------------------------------------------|
| 1    | 2B195             | Brake master cylinder and reservoir assembly                   |
| 2    | W520011           | Brake master cylinder nut (2 required)                         |
| 3    | -                 | Brake fluid level switch electrical connector (part of 14A290) |
| 4    | 2A335 RH/ 2286 LH | Master cylinder brake tube fittings                            |

#### REMOVAL AND INSTALLATION

**WARNING:** Do not use any fluid other than clean brake fluid meeting manufacturer's specification. Additionally, do not use brake fluid that has been previously drained. Following these instructions will help prevent system contamination, brake component damage and the risk of serious personal injury.

**WARNING:** Carefully read cautionary information on product label. For EMERGENCY MEDICAL INFORMATION seek medical advice. In the USA or Canada on Ford/Motorcraft products call: 1-800-959-3673. For additional information, consult the product Material Safety Data Sheet (MSDS) if available. Failure to follow these instructions may result in serious personal injury.

**CAUTION: Brake fluid is harmful to painted and plastic surfaces. If brake fluid is spilled onto a painted or plastic surface, immediately wash it with water.**

1. Remove the air cleaner and air cleaner outlet tube. For additional information, refer to **INTAKE AIR DISTRIBUTION AND FILTERING** article.
2. Remove the battery and battery tray. For additional information, refer to **BATTERY, MOUNTING AND CABLES** article.
3. Disconnect the brake fluid level switch electrical connector.
4. Using a suitable suction device, remove the brake fluid from the brake master cylinder reservoir.
5. Disconnect the master cylinder brake tube fittings, plug the brake tubes and the brake master cylinder ports.
  - To install, tighten to 28 Nm (21 lb-ft).
6. Remove the 2 brake master cylinder nuts.
  - To install, tighten to 25 Nm (18 lb-ft).
7. Remove the brake master cylinder and reservoir.
8. To install, reverse the removal procedure.
  - Bleed the master cylinder and brake system. For additional information, refer to **BRAKE SYSTEM - GENERAL INFORMATION** article.